

Mawlana Bhashani Science and Technology University
Department of Information and Communication Technology
4th year 1st Semester B. Sc (Engg.) Final Examination 2024

Course Title: E-Commerce and Web Programming
Full Marks: 70

Course Code: ICT 4107
Time: 3 Hours

Answer any five from the following questions

1. (a) What is E-Commerce? Which technologies are commonly used to build E-Commerce websites (frontend and backend)? 3
(b) What are the main components of a typical e-commerce system architecture? Explain the role of a payment gateway in e-commerce transactions. 4
(c) What is the main concept of E-Business model? Compare the technical differences between a pure-play e-commerce business model and a bricks-and-clicks model. 3
(d) Explain the Business-to-Consumer (B2C) E-commerce model in detail. Explain how inventory management is handled in a multi-vendor e-commerce platform. 4
- 2/ (a) What is Search Engine Optimization (SEO) in the context of E-commerce programming, and how does it help improve the visibility, traffic, and ranking of online stores in search engine results through optimized web development practices? 4
(b) What is Access network? Discuss main five (05) different types of access network. 4
(c) What is VoIP? Describe its working procedure in detail, and support your explanation with a suitable network architecture diagram and a VoIP protocol stack diagram to illustrate how voice data is transmitted over IP networks. 6
- 3/ (a) What is Web 2.0. Discuss its key features with a real-world example that demonstrates its impact on modern web applications. 3
(b) Draw and explain the layers of the Internet model in sequential order, and discuss how each layer functions collaboratively to enable the sending and receiving of data across a network. 4
(c) Discuss the Three-Tier Application Architecture and describe the relevant programming languages, tools, and technologies commonly used in each layer for building scalable and maintainable web applications. 4 
(d) Describe Internetworking and Intranetworking with proper protocol and example. 3
- 4/ (a) How are the <a>, , <p>, and <div> tags used in HTML to structure and link content on a web page? Provide examples to demonstrate their usage and explain their roles in webpage layout and navigation. 4
(b) HTML's <input> element supports a variety of input types, each rendering differently in the browser and serving specific user interaction needs. Explore and list the different type attributes that can be applied to the <input> element, briefly explaining how it is typically used in a web form. 4
(c) Create a 4-row by 6-column HTML table where the first cell in the first row stretches across two columns. The third cell in the same row extends further to cover the remaining four columns in that row. In the third row, the third cell is extended to cover the space of two rows. All remaining cells should follow the regular structure. Use your own data to fill the table. Apply CSS to display clear borders around each cell, center all text in the middle of the cells, add padding for readability, and highlight any extended cells with a soft background color to distinguish them. 6
- 5/ (a) What is CSS in the context of web development? In what scenarios would a web developer prefer internal CSS over external CSS? Justify with examples. 3
(b) A developer is working on a team project where multiple pages share the same layout and styles. Which CSS method would be most appropriate and why? Discuss methods with clear explanations and relevant coding examples for each. 4
(c) Design a responsive horizontal navigation bar using HTML and CSS, where the menu includes multiple top-level items such as Home, Profile, Skill, Project, and Contact. The Profile and Skill menu items should each contain a dropdown submenu that appears when the user hovers over them. The Profile submenu must include Educational Profile, Technical Profile, CV, and Video Resume, while the Skill submenu includes General Skill and Technical Skill. Style the navigation bar using a yellow background, blue link text, and rounded gray hover effects. Ensure that the dropdown menus are styled with matching background colors and appear directly below their parent items on hover. Implement all styling using an external CSS file, and apply clear layout management using box-sizing and font control for a clean user interface. 7

- 6/ (a) "JavaScript can update and change both HTML and CSS, it can also calculate, manipulate, and validate data" – Justify the statement with coding. 4
- (b) Design a simple interactive counter application using HTML, CSS, and JavaScript. Your task is to display a numeric counter on the webpage, initially set to zero, and provide three buttons labeled **Increase**, **Reset**, and **Decrease**. When the user clicks the Increase button, the counter should increment by one. Clicking the Decrease button should decrement the counter by one, and the Reset button should set the counter back to zero. Style the layout so that the counter number is prominently displayed at the center with a large font size. Position the buttons below the counter, evenly spaced, and styled with appropriate padding and font size for better user interaction. Ensure that the counter updates instantly on each button click without reloading the page. 6
- (c) In what scenarios would using synchronous code be preferred over asynchronous code in a JavaScript application? Justify your answer with proper coding. 4
7. (a) What is API? At which OSI layer does SSL/TLS encryption operate, and why is it important for online payments? 3
- (b) You are tasked with developing a JavaScript-based feature for a web application that verifies the validity and deliverability of a user's email address before allowing them to proceed with registration. To accomplish this, you will use the Abstract Email Validation API, which provides detailed information about an email address, including its deliverability status. Write a JavaScript program that uses the Fetch API to send a GET request to the API endpoint <https://emailvalidation.abstractapi.com/v1/>, passing the API key `1f38e6ff3cae4458bf885301abc5a21f` and the email address `testmail@gmail.com` as query parameters. The program should then parse the JSON response, check the deliverability field, and print "The EMAIL is deliverable" if the field's value is 'DELIVERABLE'. Otherwise, it should print "The EMAIL is NOT deliverable." 5
- (c) What is JSON? In case of a DoS attack on an e-commerce server, which TCP/IP layer is most likely targeted and why? 3
- (d) How is Bluetooth technology used in e-commerce, particularly in retail environments? Describe how 4G contributes to real-time inventory updates and location-based services in mobile shopping. 3
8. (a) Write answer-based on web server: 6
- (i) Describe web server with working procedure of web server. [2]
- (ii) How does load balancing work with web servers in an e-commerce environment? [3]
- (iii) Mention which web development stack Apache belongs to? [1]
- (b) "PHP (Hypertext Preprocessor) is a general-purpose scripting language that is mainly used for server-side scripting." Justify this statement by creating a form where the user inputs data, and the server stores or processes it using PHP. Include both the HTML form and the PHP processing script in your answer. 5
- (c) Write short answer: 3
- (i) What is CGI (Common Gateway Interface) in web development?
- (ii) Mention the types of programming languages typically used to write CGI scripts?
- (iii) Differentiate RESTful APIs and SOAP APIs.